APICAL HELSON–SZEGŐ WEIGHTS AND SATURATED OUTER FUNCTIONS

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ABSTRACT. A function φ in H^∞ is called saturated if the Hankel operator with symbol φ has norm $\|\varphi\|_\infty$. Brevig and Seip showed that an outer function φ is not saturated if, at some level, its modulus is an apical Helson–Szegő weight and the corresponding sublevel set has nonempty interior. They asked whether apical Helson–Szegő weights are invariant under positive powers, and which, if any, of the two conditions are necessary. We show that the answer to the first question is affirmative: any finite bound for the conjugate function on the top set can be made arbitrarily small. For the second question, we show that neither condition is necessary. The interior condition can be removed altogether, and we construct a bounded outer function which is not saturated although its modulus is not an apical Helson–Szegő weight at any level. The description of the saturated outer functions remains open.

1. INTRODUCTION

Let $\mathbb{T}$ be the unit circle, equipped with normalized arc length measure m, and let H^∞ denote the space of bounded analytic functions in the unit disc $\mathbb{D}$, identified with their boundary values on $\mathbb{T}$ and equipped with the norm

$$\|\varphi\|_\infty = \operatorname*{ess\,sup}_{\zeta\in\mathbb{T}} |\varphi(\zeta)|.$$

Let H^∞_0 be the subspace of functions vanishing at the origin. For a symbol $\varphi \in H^\infty$, the Hankel operator H_φ considered by Brevig and Seip [1, equations (1) and (2)] satisfies $\|\varphi\|_2 \le \|H_\varphi\| \le \|\varphi\|_\infty$, and by Nehari's theorem [6] its norm is the distance from φ to the complex conjugates of H^∞_0. Following [1, equation (4)], we say that φ is *saturated* if

$$\|\varphi\|_\infty = \inf_{f\in H^\infty_0} \|\varphi - \overline{f}\|_\infty,$$

that is, if H_φ has maximal norm. Since $\|\varphi - \overline{f}\|_\infty = \|\overline{z}\,\overline{\varphi} - \overline{z}f\|_\infty$ and $\overline{z}f$ ranges over H^∞ as f ranges over H^∞_0, φ is saturated if and only if $\overline{z}\,\overline{\varphi}$ is badly approximable, that is, its distance to H^∞ equals its norm; see Peller [7] for this classical notion. Functions of constant modulus are saturated, and Brevig and Seip [1, Theorem 3] described the saturated functions with continuous

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but nonconstant modulus. In general the problem is open, and the results of [1] indicate that saturation is tied to irregular behavior of $\log|\varphi|$ or of the argument of φ near the points where $|\varphi|$ is close to its maximum.

To make this precise, Brevig and Seip use the super-level and sub-level sets of a bounded function w,

$$L^+_w(\delta) = \{\zeta \in \mathbb{T} : |w(\zeta)| \geq \delta\|w\|_\infty\}, \quad L^-_w(\delta) = \{\zeta \in \mathbb{T} : |w(\zeta)| < \delta\|w\|_\infty\},$$

for $0 \leq \delta \leq 1$. We call $L^+_w(\delta)$ a *top set* of w. Recall that a positive function w is a Helson–Szegő weight if $w = \exp(u + \widetilde{v})$ with real $u, v \in L^\infty$ and $\|v\|_\infty < \pi/2$, where $\widetilde{v}$ denotes the conjugate function of v. By the theorem of Helson and Szegő [4] these are the weights w for which the Riesz projection is bounded on $L^2(w)$, and by the theorem of Hunt, Muckenhoupt and Wheeden [5] they are the (A_2) weights. Only the behavior of w near its maximum matters for saturation, and Brevig and Seip [1, Section 1] made the following definition: a bounded function w is an *apical Helson–Szegő weight* if there exists some $0 < \delta < 1$ such that

$$w = \exp(u + v), \tag{1}$$

where u is in L^∞, v is in L^1, and

$$\operatorname*{ess\,sup}_{L^+_w(\delta)} |\widetilde{v}| < \frac{\pi}{2}. \tag{2}$$

Here u and v are real-valued. When we wish to specify δ, we say that w is apical at level δ. In [1] the conditions (1) and (2) are numbered (6) and (7). For an outer function φ, the inner factor is trivial and its spectrum is empty, and the main sufficient condition for nonsaturation [1, Theorem 5] reads as follows: *an outer function $\varphi \in H^\infty$ is not saturated if there is some $0 < \delta < 1$ such that $|\varphi|$ is an apical Helson–Szegő weight at level δ and the interior of $L^-_\varphi(\delta)$ is nonempty.*

At the end of their paper, Brevig and Seip posed two problems.

Problem (Brevig–Seip [1, Problem 1])**.** *Describe the outer functions in H^∞ that are saturated.*

They continue: "By Theorem 4,[1] an outer function φ fails to be saturated if there is a $0 < \delta < 1$ such that $|\varphi|$ is an apical Helson–Szegő weight at level δ and the interior of $L^-_\varphi(\delta)$ is nonempty. It is natural to ask which, if any, of these two conditions may also be necessary for φ not to be saturated."

Problem (Brevig–Seip [1, Problem 2])**.** *Is it true that w is an apical Helson–Szegő weight if and only if w^κ is an apical Helson–Szegő weight for all positive numbers κ?*

They add: "In other words, we ask if (7) could be replaced simply by (27)", where (27) is the condition

$$\operatorname*{ess\,sup}_{L^+_w(\delta)} |\widetilde{v}| < \infty. \tag{3}$$

[1]Theorem 5 of [1] appears to be meant here.

Brevig and Seip motivate Problem 2 of [1] by a family of weights w_κ, built from a thin open subset of $\mathbb{T}$, which are Helson–Szegő weights only for $0 < \kappa < 1$, but apical at level $1/2$ for every $\kappa > 0$.

The main purpose of the present paper is to answer Problem 2 of [1] and the question following Problem 1 of [1]. We begin with the former.

Theorem 1. *Suppose that w is a bounded positive function on $\mathbb{T}$, that $0 < \delta < 1$, and that $\log w = u + v$ with real $u \in L^\infty$ and $v \in L^1$ such that (3) holds. Then for every $\eta > 0$ there are real $u_\eta \in L^\infty$ and $v_\eta \in L^1$ with $\log w = u_\eta + v_\eta$ and*

$$\operatorname*{ess\,sup}_{L^+_w(\delta)} |\widetilde{v_\eta}| < \eta.$$

In particular, w is an apical Helson–Szegő weight at level δ if and only if w^κ is an apical Helson–Szegő weight at level δ^κ, for every $\kappa > 0$.

Hence the answer to Problem 2 of [1] is affirmative, and condition (2) may be replaced by (3) at the same level δ. Brevig and Seip noted [1, Section 3] that this would imply that w is an apical Helson–Szegő weight whenever $\log w$ has bounded mean oscillation; see Corollary 1 below.

We turn to the question following Problem 1 of [1]. Our second result shows that the interior condition is not needed at all.

Theorem 2. *Suppose that $\varphi \in H^\infty$ is an outer function whose modulus is not almost everywhere constant, and that there are $0 < \delta < 1$, real $u \in L^\infty$ and $v \in L^1$ with $\log|\varphi| = u + v$ and*

$$\operatorname*{ess\,sup}_{L^+_\varphi(\delta)} |\widetilde{v}| < \infty.$$

Then φ is not saturated.

In particular, an outer function whose modulus is a nonconstant apical Helson–Szegő weight is not saturated. If $A \subseteq \mathbb{T}$ is a closed nowhere dense set with $0 < m(A) < 1$ and φ is the outer function with $|\varphi| = 1/2$ on A and $|\varphi| = 1$ elsewhere, then $L^-_\varphi(\delta)$ has empty interior for every $0 < \delta < 1$, while Theorem 2 applies with $u = \log|\varphi|$ and $v = 0$. Hence the interior condition is not necessary.

Our third result shows that the apical condition is not necessary either, not even in the weak form of Theorem 2.

Theorem 3. *There is an outer function $\varphi \in H^\infty$ with $\|\varphi\|_\infty = 1$ such that*

$$\inf_{f \in H^\infty_0} \|\varphi - \overline{f}\|_\infty \leq \sqrt{575/576},$$

but such that for no $0 < \delta < 1$ can one write $\log|\varphi| = u + v$ with real $u \in L^\infty$ and $v \in L^1$ and with $\widetilde{v}$ essentially bounded on $L^+_\varphi(\delta)$. In particular, φ is not saturated, but $|\varphi|$ is not an apical Helson–Szegő weight at any level.

Theorems 2 and 3 together show that neither of the two conditions of [1, Theorem 5] is necessary for an outer function to be nonsaturated. They do

not answer Problem 1 of [1], which remains open; we return to it in Section 6. To the best of our knowledge, Theorems 1 to 3 are new.

Let us briefly describe our approach. For Theorem 1 we may assume that $v \leq 0$. Then the analytic function F with boundary values $v + i\widetilde{v}$ takes values in the left half-plane, and we move the bounded analytic function

$$B_N = \frac{NF}{N - F}$$

from v to u. Since B_N is analytic, its conjugate comes along for free, and $F - B_N$ is of size $|F|^2/N$ on the top set. In contrast to the truncation $\max(\log w, -R)$ in the proof of [1, Lemma 7], which cuts the real part and then needs a separation condition to control the conjugate, we truncate F itself. For Theorem 2 we observe that in the proof of [1, Theorem 5] the interior of $L^-_\varphi(\delta)$ is used only to compensate for the argument of the factor z which makes the competitor vanish at the origin. We replace z by a function $k \in H^\infty_0$ which is close to 1 on the top set, and this requires only that the sublevel set has positive measure. For Theorem 3 we take an outer function F whose argument V tends to $-\infty$ at a point, and divide by a bounded outer function so that the top set becomes the set E where $\cos V \geq 1/2$. Nonsaturation sees only the direction of φ on the top set, that is, V modulo 2π, while apicality asks for a bounded continuous branch of the argument there, up to the conjugate of a bounded function. We arrange that V is not square integrable on E, whereas conjugates of bounded functions are.

Organization. The present paper is organized into four additional sections. In Section 2 we fix notation for conjugate functions and outer functions. In Section 3 we prove Theorem 1 and deduce the consequence for weights of bounded mean oscillation. In Section 4 we give a criterion for nonsaturation and prove Theorem 2. In Section 5 we prove Theorem 3. Section 6 contains concluding remarks and a problem.

2. Preliminaries

For a real function $q \in L^1$, let

$$\mathcal{H}(q)(z) = \int_{\mathbb{T}} \frac{\zeta + z}{\zeta - z}\, q(\zeta)\, \mathrm{d}m(\zeta), \qquad z \in \mathbb{D},$$

be its Herglotz integral. The real part of $\mathcal{H}(q)$ is the Poisson integral of q, and $\mathcal{H}(q)(0) = \int_{\mathbb{T}} q\, \mathrm{d}m$ is real. The conjugate function $\widetilde{q}$ is defined by the almost everywhere boundary values $\mathcal{H}(q) = q + i\widetilde{q}$, so the harmonic extension of $\widetilde{q}$ vanishes at the origin. In angular coordinates [3, Chapter III],

$$\widetilde{q}(e^{it}) = \lim_{\varepsilon \to 0^+} \int_{\varepsilon < |t-\theta| \leq \pi} \cot \frac{t - \theta}{2}\, q(e^{i\theta}) \frac{\mathrm{d}\theta}{2\pi} \tag{4}$$

for almost every t; this is the formula used in the proof of [1, Lemma 7]. Conjugation is linear. If $B \in H^\infty$ and $B(0)$ is real, then the conjugate of

$\operatorname{Re} B$ is $\operatorname{Im} B$. On Fourier coefficients, conjugation multiplies the coefficient of index n by $-i\operatorname{sgn}(n)$, so by Parseval's identity $\|\widetilde{q}\|_2 \le \|q\|_2$. In particular,

$$q \in L^\infty \implies \widetilde{q} \in L^2. \tag{5}$$

If $q \in L^1$ is real and bounded above, then $\exp \mathcal{H}(q)$ is the outer function with modulus e^q on $\mathbb{T}$. Since the Poisson kernel is positive, $|\exp \mathcal{H}(q)| \le \exp(\operatorname{ess\,sup} q)$ in $\mathbb{D}$. Every outer function $\varphi \in H^\infty$ is of the form $\varphi = \gamma \exp \mathcal{H}(\log|\varphi|)$ for a unimodular constant γ; this is [1, equation (13)], see also [3, Section II.5]. In particular, if $\log|\varphi| = u + v$ with $u \in L^\infty$, then

$$\varphi = \gamma \exp \mathcal{H}(u) \exp \mathcal{H}(v), \tag{6}$$

and $\exp \mathcal{H}(u) = \exp(u + i\widetilde{u})$ is invertible in H^∞, with $|\exp \mathcal{H}(u)|$ between $e^{-\|u\|_\infty}$ and $e^{\|u\|_\infty}$ in $\mathbb{D}$. The boundary values of $\exp \mathcal{H}(v)$ are $e^{v+i\widetilde{v}}$, so $\widetilde{v}$ is a continuous choice of the argument of $\varphi/(\gamma \exp \mathcal{H}(u))$ on $\mathbb{T}$, in the sense that no multiple of 2π has been discarded. Condition (2) asks that this argument be small on the top set, and (3) that it be bounded there. In the notation of [1, Section 2], (6) reads $\varphi = \gamma \exp(u + i\widetilde{u} + v + i\widetilde{v})$.

3. Power invariance of apical weights

The idea of the proof of Theorem 1 is that a finite bound on the top set can be shrunk by moving a bounded analytic function from v to u, without changing the top set.

Proof of Theorem 1. Write $E = L^+_w(\delta)$. Since $\log w$ and u are bounded above, so is v. Moving a real constant from v to u, we may therefore assume that $v \le 0$. This does not change $\widetilde{v}$, since the conjugate of a constant is 0. Let $F = \mathcal{H}(v)$. Then $\operatorname{Re} F \le 0$ in $\mathbb{D}$, $F(0)$ is real, and the boundary values of F are $v + i\widetilde{v}$ almost everywhere.

On E we have $\log(\delta\|w\|_\infty) \le \log w \le \log\|w\|_\infty$, so $v = \log w - u$ is essentially bounded on E. By (3) so is $\widetilde{v}$. Hence there is a finite K such that $|F| \le K$ almost everywhere on E.

For $N > 0$ we set

$$B_N = \frac{NF}{N - F}.$$

The Möbius map $z \mapsto Nz/(N - z)$ maps the left half-plane into the disc with diameter $[-N, 0]$. To verify what we need, note that since $\operatorname{Re} F \le 0$ we have $|N - F| \ge N$ and $|N - F|^2 - |F|^2 = N^2 - 2N \operatorname{Re} F \ge N^2$. Hence B_N is analytic in $\mathbb{D}$ with $|B_N| \le N$, and $B_N(0)$ is real. We set

$$u_N = u + \operatorname{Re} B_N \qquad \text{and} \qquad v_N = v - \operatorname{Re} B_N,$$

using boundary values. Then $u_N \in L^\infty$, $v_N \in L^1$ and $\log w = u_N + v_N$. Since $B_N(0)$ is real, the conjugate of $\operatorname{Re} B_N$ is $\operatorname{Im} B_N$, and so $\widetilde{v_N} = \operatorname{Im}(F - B_N)$. A direct computation gives

$$F - B_N = -\frac{F^2}{N - F}.$$

Hence, almost everywhere on E,

$$|\widetilde{v_N}| \le |F - B_N| = \frac{|F|^2}{|N - F|} \le \frac{K^2}{N}.$$

Choosing $N > K^2/\eta$, we obtain the first assertion with $u_\eta = u_N$ and $v_\eta = v_N$.

For the second assertion, note that $\|w^\kappa\|_\infty = \|w\|_\infty^\kappa$, and so

$$L^+_{w^\kappa}(\delta^\kappa) = L^+_w(\delta).$$

Suppose that w is apical at level δ, with $\log w = u + v$. Then $\log w^\kappa = \kappa u + \kappa v$, and $|\kappa\widetilde{v}| < \kappa\pi/2$ on $L^+_{w^\kappa}(\delta^\kappa)$. This is condition (3) for w^κ at level δ^κ, and the first assertion with $\eta = \pi/2$ shows that w^κ is apical at level δ^κ. The converse follows by applying the same argument to w^κ and the exponent $1/\kappa$. □

The analytic truncation in the proof leaves v unchanged where $|F|$ is much smaller than N, and changes it only where v is very negative or $\widetilde{v}$ is very large, which happens off the top set. We next record the consequence mentioned by Brevig and Seip [1, Section 3].

Corollary 1. *Suppose that w is a bounded positive function on $\mathbb{T}$ and that $\log w$ has bounded mean oscillation. Then w is an apical Helson–Szegő weight at every level $0 < \delta < 1$.*

Proof. By the theorem of Fefferman and Stein [2] (see also [3, Chapter VI]), there are real $u, q \in L^\infty$ with $\log w = u + \widetilde{q}$. Set $v = \widetilde{q}$, which is in L^1. Since $\widetilde{\widetilde{q}} = -q + \int_{\mathbb{T}} q \, \mathrm{d}m$, the conjugate of v is bounded on all of $\mathbb{T}$. Hence (3) holds at every level, and Theorem 1 applies. □

4. A criterion for nonsaturation and the interior condition

Our next lemma describes nonsaturation by a first order perturbation. Expanding the square, we find that

$$|\varphi - \overline{f}|^2 = |\varphi|^2 - 2\,\mathrm{Re}(\varphi f) + |f|^2 \tag{7}$$

almost everywhere on $\mathbb{T}$, for $\varphi, f \in H^\infty$. If $f = tg$ for small $t > 0$, the middle term dominates. Hence $\|\varphi - t\overline{g}\|_\infty < \|\varphi\|_\infty$ if $\mathrm{Re}(\varphi g)$ is bounded below by a positive constant on a top set of φ, since off the top set there is room to spare. The lemma also shows that the requirement $g(0) = 0$ costs nothing once $|\varphi|$ is nonconstant; this is the step which replaces the interior condition of [1, Theorem 5].

Lemma 1. *Suppose that $\varphi \in H^\infty$ and $\|\varphi\|_\infty = 1$. Then φ is not saturated if and only if there are $0 < \delta < 1$, $c > 0$ and $g \in H^\infty_0$ such that*

$$\mathrm{Re}(\varphi g) \ge c \qquad \text{almost everywhere on } L^+_\varphi(\delta). \tag{8}$$

If in addition $|\varphi|$ is not almost everywhere constant, then φ is not saturated if and only if (8) holds for some $0 < \delta < 1$, $c > 0$ and $G \in H^\infty$ in place of g.

Proof. Suppose first that φ is not saturated. Then there are $f \in H_0^\infty$ and $0 < \varepsilon < 1$ such that $\|\varphi - \overline{f}\|_\infty \le 1 - \varepsilon$. Let $\delta = (1-\varepsilon)^{1/2}$. On $L_\varphi^+(\delta)$ we have $|\varphi|^2 \ge 1 - \varepsilon$, and by (7)

$$2\operatorname{Re}(\varphi f) \ge |\varphi|^2 - |\varphi - \overline{f}|^2 \ge (1-\varepsilon) - (1-\varepsilon)^2 = \varepsilon(1-\varepsilon).$$

Hence (8) holds with $g = f$.

Conversely, suppose that $g \in H_0^\infty$ satisfies (8), and let $M = \|g\|_\infty$. For $t > 0$, the identity (7) with $f = tg$ gives

$$|\varphi - t\overline{g}|^2 \le \begin{cases} 1 - 2tc + t^2M^2 & \text{on } L_\varphi^+(\delta), \\ \delta^2 + 2tM + t^2M^2 & \text{on } L_\varphi^-(\delta). \end{cases}$$

Both bounds are less than 1 for small $t > 0$. Hence φ is not saturated.

It remains to show that if $|\varphi|$ is not almost everywhere constant and $G \in H^\infty$ satisfies (8) at level δ, then some $g \in H_0^\infty$ satisfies (8) at a possibly larger level. Since $|\varphi| < 1$ on a set of positive measure, there is $\delta \le \delta' < 1$ such that $A = L_\varphi^-(\delta')$ has positive measure. Let $E = L_\varphi^+(\delta')$, which is contained in $L_\varphi^+(\delta)$ and has positive measure since $\|\varphi\|_\infty = 1$. For $R > 0$, let s be the function equal to $-R$ on E and to $Rm(E)/m(A)$ on A. Then s is bounded and $\int_{\mathbb{T}} s \,\mathrm{d}m = 0$. Let $h = \mathcal{H}(s)$, so that $h(0) = 0$ and $\operatorname{Re} h \le Rm(E)/m(A)$ in $\mathbb{D}$. The function

$$k = 1 - e^h$$

is in H_0^∞, and $|1 - k| = e^s = e^{-R}$ almost everywhere on E. Set $g = Gk$. Then $g \in H_0^\infty$, and almost everywhere on E,

$$\operatorname{Re}(\varphi g) \ge \operatorname{Re}(\varphi G) - |\varphi G||1-k| \ge c - \|G\|_\infty e^{-R}.$$

For R large, this is at least $c/2$, which is (8) at level δ'. □

The function k plays the role of the factor z in the proof of [1, Theorem 5]: it vanishes at the origin, so that $Gk \in H_0^\infty$. The difference is that k is close to 1 on the top set, so it does not rotate φG there. The price is paid on the set A, where k may be large, but there $|\varphi|$ is small and the factor t in the second part of the proof absorbs it. The constant functions show that the hypothesis on $|\varphi|$ is needed: $\varphi = 1$ and $G = 1$ satisfy (8), but $\varphi = 1$ is saturated.

Armed with Lemma 1 and Theorem 1, we can prove Theorem 2.

Proof of Theorem 2. Multiplying φ by a positive constant changes u by a constant, so we may assume that $\|\varphi\|_\infty = 1$. Let $E = L_\varphi^+(\delta)$. By Theorem 1 with $w = |\varphi|$ and $\eta = \pi/4$, we may replace u and v by other decompositions, and so assume that $|\widetilde{v}| < \pi/4$ almost everywhere on E. By (6), $\varphi = \gamma \exp \mathcal{H}(u) \exp \mathcal{H}(v)$. Set

$$G = \frac{\overline{\gamma}}{\exp \mathcal{H}(u)},$$

which is in H^∞ since u is bounded. Then $\varphi G = \exp \mathcal{H}(v)$, whose boundary values are $e^{v+i\widetilde{v}}$. On E we have $v = \log|\varphi| - u \geq \log\delta - \|u\|_\infty$. Hence, almost everywhere on E,

$$\operatorname{Re}(\varphi G) = e^v \cos\widetilde{v} \geq \delta e^{-\|u\|_\infty} \cos\frac{\pi}{4} > 0.$$

By the second part of Lemma 1, φ is not saturated. □

Combining Theorem 2 and Corollary 1, we obtain a sufficient condition in terms of the modulus alone.

Corollary 2. *Suppose that $\varphi \in H^\infty$ is an outer function whose modulus is not almost everywhere constant, and that $\log|\varphi|$ has bounded mean oscillation. Then φ is not saturated. In particular, every nonconstant outer function φ which is invertible in H^∞ is not saturated.*

For an invertible φ the corollary is also immediate from Lemma 1 with $G = 1/\varphi$. The example after Theorem 2, with $|\varphi| = 1/2$ on a closed nowhere dense set A of positive measure, is of this kind. Since $\mathbb{T} \setminus A$ is open and dense, no arc is contained in A up to a null set, so $L^-_\varphi(\delta)$ has empty interior for every $0 < \delta < 1$ and every choice of representative of $|\varphi|$, and [1, Theorem 5] does not apply. Similarly, for $1/2 < \varepsilon < \delta < 1$ the sets $L^+_\varphi(\delta) = \mathbb{T} \setminus A$ and $L^-_\varphi(\varepsilon) = A$ are at distance 0, so the separation condition of [1, Lemma 7] fails at every level where $L^-_\varphi(\varepsilon)$ is nonempty.

5. A nonsaturated outer function whose modulus is not apical

By Lemma 1, nonsaturation only requires φG to point into a sector around the positive real axis on a top set, for some bounded analytic G. In the proof of Theorem 2, apicality produces such a G with the additional property that the continuous argument $\widetilde{v}$ of φG stays in the sector. We now show that the continuous argument may wind infinitely many times on the top set, provided it returns to the sector often enough.

We work in angular coordinates $-\pi < t \leq \pi$ and set $a = \pi/8$. Define

$$v(e^{it}) = \begin{cases} -(-t)^{-1/2}, & -a < t < 0, \\ 0, & \text{otherwise.} \end{cases}$$

Then $v \leq 0$, and $v \in L^1$ since $\int_0^a x^{-1/2}\,\mathrm{d}x = 2a^{1/2}$. Let $V(t) = \widetilde{v}(e^{it})$ for $0 < t < a$. For such t the integral in (4) is absolutely convergent, since v vanishes near e^{it}, and the substitution $\theta = -x$ gives

$$V(t) = -\frac{1}{2\pi}\int_0^a x^{-1/2} \cot\frac{t+x}{2}\,\mathrm{d}x. \tag{9}$$

Thus V is smooth on $(0, a)$. Since v is a negative mass just to the left of $t = 0$, its conjugate is very negative just to the right of $t = 0$. The next lemma makes this precise.

Lemma 2. *As $t \to 0^+$,*

$$V(t) = -t^{-1/2} + O(1) \qquad \text{and} \qquad V'(t) = \frac{1}{2}t^{-3/2} + O(1).$$

Proof. For $0 < y \le 2a = \pi/4$ we have $\cot(y/2) = 2/y + O(y)$ and $\csc^2(y/2) = 4/y^2 + O(1)$. Inserting the first expansion in (9), the error term contributes at most a constant times $\int_0^a x^{-1/2}(t+x)\,\mathrm{d}x$, which is bounded. Hence $V(t)$ equals minus the left-hand side of the next display, up to a bounded error. The substitution $x = ty^2$ gives

$$\frac{1}{\pi}\int_0^a \frac{x^{-1/2}}{t+x}\,\mathrm{d}x = \frac{2}{\pi t^{1/2}}\arctan\sqrt{a/t} = t^{-1/2} - \frac{2}{\pi t^{1/2}}\arctan\sqrt{t/a},$$

and the last term is bounded since $\arctan y \le y$. This proves the first formula. Differentiating (9) under the integral sign, we get

$$V'(t) = \frac{1}{4\pi}\int_0^a x^{-1/2}\csc^2\frac{t+x}{2}\,\mathrm{d}x = \frac{1}{\pi}\int_0^a \frac{x^{-1/2}}{(t+x)^2}\,\mathrm{d}x + O(1).$$

Extending the last integral to $(0,\infty)$ costs at most $\int_a^\infty x^{-5/2}\,\mathrm{d}x$. The same substitution gives

$$\int_0^\infty \frac{x^{-1/2}}{(t+x)^2}\,\mathrm{d}x = 2t^{-3/2}\int_0^\infty \frac{\mathrm{d}y}{(1+y^2)^2} = \frac{\pi}{2}t^{-3/2},$$

which proves the second formula. □

Hence the function $F = \exp\mathcal{H}(v)$, which has boundary values $e^{iV(t)}$ for $0 < t < a$, winds around the origin infinitely many times as $t \to 0^+$, clockwise as t decreases. Let

$$E = \{e^{it} : 0 < t < a,\ \cos V(t) \ge 1/2\}$$

be the set where F points into the sector $|\arg F| \le \pi/3$. We use E also for the corresponding set of t.

Lemma 3. *The set E has positive measure, and V is not square integrable on E.*

Proof. By Lemma 2 there is $0 < t_0 < a$ such that for $0 < t < t_0$,

$$\frac{1}{2}t^{-1/2} \le |V(t)| \le \frac{3}{2}t^{-1/2} \qquad \text{and} \qquad 0 < V'(t) \le \frac{3}{4}t^{-3/2}.$$

In particular, V is an increasing diffeomorphism from $(0,t_0)$ onto $(-\infty, V(t_0))$. Let $S = \{y \le V(t_0) : \cos y \ge 1/2\}$. We change variables $y = V(t)$. Since

$$\frac{V(t)^2}{V'(t)} \ge \frac{t^{-1}/4}{3t^{-3/2}/4} = \frac{t^{1/2}}{3} \ge \frac{1}{6|V(t)|},$$

we find that

$$\int_{E\cap(0,t_0)} V(t)^2\,\mathrm{d}t = \int_S \frac{V(t)^2}{V'(t)}\,\mathrm{d}y \ge \frac{1}{6}\int_S \frac{\mathrm{d}y}{|y|},$$

where $t = V^{-1}(y)$ in the middle integral. The set S contains the interval $[2\pi j - \pi/3, 2\pi j + \pi/3]$ for every sufficiently negative integer j, so the last integral diverges like the harmonic series. Similarly, $E \cap (0, t_0)$ has measure $\int_S \mathrm{d}y/V'(V^{-1}(y)) > 0$. □

In the variable $y = V(t)$, the top set E occupies a fixed proportion of every period, while $V^2\,\mathrm{d}t$ is comparable to $\mathrm{d}y/|y|$. This is why the continuous argument is not square integrable on E.

Proof of Theorem 3. Let $s = (\log 2)\mathbf{1}_{\mathbb{T}\setminus E}$, and define the outer functions

$$F = \exp \mathcal{H}(v), \qquad G = \exp \mathcal{H}(s) \qquad \text{and} \qquad \varphi = \frac{F}{G} = \exp \mathcal{H}(v - s).$$

Since $v \le 0$ and $0 \le s \le \log 2$, the Poisson integral gives $|F| \le 1$ and $1 \le |G| \le 2$ in $\mathbb{D}$. Hence G is invertible in H^∞ and $|\varphi| \le 1$. On E we have $v = s = 0$, so $|\varphi| = 1$, while off E we have $|\varphi| = e^{v - \log 2} \le 1/2$. By Lemma 3, E has positive measure. Hence $\|\varphi\|_\infty = 1$, the modulus of φ is not constant, $E \subseteq L^+_\varphi(\delta)$ for every $0 < \delta < 1$, and $L^+_\varphi(\delta) = E$ up to a null set for $1/2 < \delta < 1$. Moreover, on E the boundary values of $\varphi G = F$ are e^{iV}, so

$$\operatorname{Re}(\varphi G) = \cos V \ge \frac{1}{2} \qquad \text{almost everywhere on } E. \tag{10}$$

We first prove the bound on the distance. We follow the second part of the proof of Lemma 1 with explicit constants. Since $m(E) \le a/(2\pi) = 1/16$, the set $A = \mathbb{T} \setminus E$ has $m(A) \ge 15/16$. Let $R = \log 8$, and let $h = \mathcal{H}(r)$, where $r = -R$ on E and $r = Rm(E)/m(A)$ on A. Then $h(0) = 0$, $\operatorname{Re} h \le R/15$ in $\mathbb{D}$, and $|e^h| = 1/8$ almost everywhere on E. Let $g = G(1 - e^h)$. Then $g \in H^\infty_0$ and

$$\|g\|_\infty \le 2(1 + 8^{1/15}) < 6.$$

On E we have $|F| = 1$, and so by (10)

$$\operatorname{Re}(\varphi g) = \operatorname{Re} F - \operatorname{Re}(F e^h) \ge \frac{1}{2} - \frac{1}{8} > \frac{1}{4}.$$

Let $f = g/144$, so that $\|f\|_\infty < 1/24$. By (7), almost everywhere on E

$$|\varphi - \overline{f}|^2 \le 1 - \frac{2}{4 \cdot 144} + \frac{1}{24^2} = \frac{575}{576},$$

and almost everywhere on A, where $|\varphi| \le 1/2$,

$$|\varphi - \overline{f}|^2 \le \Big(\frac{1}{2} + \frac{1}{24}\Big)^2 = \frac{169}{576}.$$

This proves the bound on the distance, and in particular φ is not saturated.

Suppose now that for some $0 < \delta < 1$ we have $\log|\varphi| = u + w$, with real $u \in L^\infty$ and $w \in L^1$, and that $\widetilde{w}$ is essentially bounded on $L^+_\varphi(\delta)$. Since $\log|\varphi| = v - s$, we have $v = w + (u + s)$, and so

$$V = \widetilde{w} + \widetilde{u + s} \qquad \text{almost everywhere on } E.$$

The first term is bounded on $E \subseteq L^+_\varphi(\delta)$, and the second is in L^2 by (5), since $u+s$ is bounded. Hence V is square integrable on E, which contradicts Lemma 3. □

In the example, $\widetilde{v} = V$ is unbounded on the top set E, but this alone would not prevent apicality, since v could be changed by a bounded function. The point of Lemma 3 is that any such change alters the conjugate only by an L^2 function, while the winding of F is too fast to be undone in L^2.

6. Concluding remarks

Saturated outer functions. For outer functions of nonconstant modulus, Theorems 2 and 3 show that the apical condition lies strictly between the conditions of [1, Theorem 5] and nonsaturation, but they do not describe the saturated outer functions, and Problem 1 of [1] remains open. Lemma 1 is a reformulation of nonsaturation, quantified over an unknown multiplier G, rather than a condition on $|\varphi|$. It does, however, suggest a candidate. Suppose that φ is outer with $\|\varphi\|_\infty = 1$. If $\log|\varphi| = u + v$ as in (1), then $v = \log|\varphi| + s$ with $s = -u$. Hence $|\varphi|$ satisfies condition (3) at level δ, and so is apical at level δ by Theorem 1, if and only if there is a real $s \in L^\infty$ such that $\widetilde{\log|\varphi|} + \widetilde{s}$ is essentially bounded on $L^+_\varphi(\delta)$. On the other hand, if $G \in H^\infty$ is invertible, then G is outer with bounded logarithmic modulus, so $G = \gamma \exp \mathcal{H}(s)$ for a real $s \in L^\infty$ and a unimodular constant γ. The boundary values of φG then have argument $\beta + \widetilde{\log|\varphi|} + \widetilde{s}$ for a real constant β, and modulus between $\delta e^{-\|s\|_\infty}$ and $e^{\|s\|_\infty}$ on $L^+_\varphi(\delta)$. Hence (8) holds for an invertible G if and only if

$$\cos\bigl(\beta + \widetilde{\log|\varphi|} + \widetilde{s}\bigr) \geq c \qquad \text{almost everywhere on } L^+_\varphi(\delta) \tag{11}$$

for some real $s \in L^\infty$, real β and $c > 0$. This is a condition on $|\varphi|$ alone. Apicality asks that the continuous argument $\widetilde{\log|\varphi|} + \widetilde{s}$ be bounded on the top set, while (11) asks only that it lie in a fixed sector modulo 2π. The proof of Theorem 2 shows that apicality implies (11), and the function in Theorem 3 satisfies (11) by (10), with $s = \log 2 \cdot \mathbf{1}_{\mathbb{T}\setminus E}$, without being apical. By Lemma 1, (11) implies nonsaturation when $|\varphi|$ is nonconstant. We do not know whether the converse holds.

Problem. *Suppose that $\varphi \in H^\infty$ is an outer function which is not saturated. Is there an invertible $G \in H^\infty$ such that $\operatorname{Re}(\varphi G) \geq c > 0$ almost everywhere on $L^+_\varphi(\delta)$ for some $0 < \delta < 1$? Equivalently, is there a real $s \in L^\infty$ such that (11) holds for some $\beta \in \mathbb{R}$, $c > 0$ and $0 < \delta < 1$?*

An affirmative answer would describe the nonsaturated outer functions in terms of their modulus. The difficulty is that Lemma 1 provides only pointwise information on a top set, while replacing G by an invertible function requires control on all of $\mathbb{T}$.

Inner factors. Theorems 2 and 3 concern outer functions. For $\varphi = \Phi I$ with a nontrivial inner factor I, condition (ii) of [1, Theorem 5] also requires the spectrum of I to lie in the interior of $L^-_\Phi(\delta)$, and there it serves to unwind the argument of I as well. Lemma 1 holds for every $\varphi \in H^\infty$, but the construction of G in the proof of Theorem 2 uses that φ is outer.

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